

M. Riley Owens

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ABOUT ME

I am a first-year graduate student in the Department of Astronomy at the University of California, Berkeley, where I work with Prof. Dan Stark. My research interests focus on understanding the evolution and physics of galaxies in the early universe, from the end of reionization to the first galaxies.

EDUCATION

Doctor of Philosophy in Astrophysics 2025 -
University of California, Berkeley

Advised by Prof. Daniel Stark | GPA: 3.87

Bachelor of Science in Astrophysics and Physics, Minor in Mathematics 2018-2022
University of Cincinnati

Advised by Prof. Matthew Bayliss | GPA: 3.73

POSITIONS

Graduate student researcher 2025 -
University of California, Berkeley

Postbaccalaureate researcher 2022
West Virginia University

Postbaccalaureate researcher 2022-2025
University of Cincinnati

NSF REU student 2021
West Virginia University

Undergraduate researcher 2019-2022
University of Cincinnati

AWARDS AND HONORS

NSF GRFP Honorable Mention, National Science Foundation 2025

Fulbright Semi-Finalist, Fulbright Program 2023

Chambliss Astronomy Achievement Award, American Astronomical Society 2023

Dean's List, University of Cincinnati 2018-2022

University Honors Scholar, University of Cincinnati 2018-2022

GRANTS, SCHOLARSHIPS, AND FELLOWSHIPS

Undergraduate Research Fellowship, University of Cincinnati \$750 2022

Cincinnatus Scholarship, University of Cincinnati \$X? 2018-2022

PUBLICATIONS

For a complete record of my publications and associated statistics, please visit [my ORCID record on NASA ADS](#).

FIRST AUTHOR

Connecting Lya and Ionizing Photon Escape in the Sunburst Arc, 5 citations 2024
The Astrophysical Journal

DOI: [10.3847/1538-4357/ad9247](https://doi.org/10.3847/1538-4357/ad9247)

M. R. Owens, K. J. Kim, M. B. Bayliss, T. E. Rivera-Thorsen, K. Sharon, J. R. Rigby, A. Navarre, M. Florian, M. D. Gladders, J. G. Burns, G. Khullar, J. Chisholm, G. Mahler, H. Dahle, C. M. Malhas, B. Welch, T. A. Hutchison, R. Gassis, S. Choe, P. Adhikari

COAUTHOR

Lyman Continuum escaping from in-situ formed stars in a tidal bridge at $z=3$, 2025
arXiv

DOI: [10.48550/arXiv.2512.19967](https://doi.org/10.48550/arXiv.2512.19967)

T. E. Rivera-Thorsen, A. Le Reste, M. J. Hayes, S. Flury, A. Saldana-Lopez, B. Welch, S. Choe, K. Sharon, K. Kim, **M. R. Owens**, E. Solhaug, H. Dahle

JWST & the Waz Arc I: Spatially Resolving the Physical Conditions within a Post-Starburst Galaxy at Redshift 5 with NIRSpec IFS, 2025
arXiv

DOI: [10.48550/arXiv.2512.02000](https://doi.org/10.48550/arXiv.2512.02000)

T. A. Hutchison, G. Khullar, J. R. Rigby, B. Welch, M. K. Florian, K. Sharon, I. Sierra, J. Sarmiento, G. Mahler, N. J. Cleri, R. Bezanson, M. D. Gladders, M. B. Bayliss, J. S. M. Karp, D. Berry, A. Ross, T. E. Rivera-Thorsen, S. C. Choe, H. Dahle, J. Chisholm, E. L. Lambrides, R. L. Larson, G. M. Olivier, **M. R. Owens**, E. Solhaug

Tracing Structure: Shape and Centroid Deviations in 39 Strong Lensing Clusters as a Test of Cluster Formation Predictions, 2025
arXiv

DOI: [10.48550/arXiv.2511.22661](https://doi.org/10.48550/arXiv.2511.22661)

R. Gassis, M. B. Bayliss, K. Sharon, G. Mahler, M. D. Gladders, M. McDonald, H. Dahle, M. K. Florian, J. R. Rigby, L. A. Elicker, **M. R. Owens**, P. Adhikari, G. Khullar

The Chandra Strong Lens Sample: Measuring the Dynamical States and Relaxation Fraction of a Sample of 28 Strong Lensing Selected Galaxy Clusters, 1 citations 2025
arXiv

DOI: [10.48550/arXiv.2511.12707](https://doi.org/10.48550/arXiv.2511.12707)

R. Gassis, M. B. Bayliss, M. McDonald, K. Sharon, G. Mahler, M. D. Gladders, H. Dahle, M. K. Florian, J. R. Rigby, L. A. Elicker, **M. R. Owens**, P. Adhikari, G. Khullar

The Sunburst Arc with JWST. IV. The importance of interaction, turbulence, and feedback for Lyman continuum escape, 2 citations 2025
arXiv

DOI: [10.48550/arXiv.2510.11702](https://doi.org/10.48550/arXiv.2510.11702)

T. E. Rivera-Thorsen, B. Welch, T. Hutchison, M. J. Hayes, J. R. Rigby, K. Kim, S. Choe, M. Florian, M. B. Bayliss, G. Khullar, K. Sharon, H. Dahle, J. Chisholm, E. Solhaug, **M. R. Owens**, M. D. Gladders

Cool Dark Gas in Cygnus X: The First Large-scale Mapping of Low-frequency Carbon Recombination Lines, 1 citations 2025
The Astrophysical Journal

DOI: [10.3847/1538-4357/adfa17](https://doi.org/10.3847/1538-4357/adfa17)

K. L. Emig, P. Salas, L. D. Anderson, D. A. Roshi, L. Bonne, A. D. Bolatto, I. A. Grenier, R. C. Levy, D. J. Linville, M. Luisi, **M. R. Owens**, J. Poojapriyatharsheni, N. Schneider, L. Tibaldo, A. G. G. M. Tielens, S. K. Walch, G. J. White

- COOL -LAMPS. VIII. Known Wide-separation Lensed Quasars and Their Host Galaxies Reveal a Lack of Evolution in M_{BH}/M_* Since $z \sim 3$,** *The Astrophysical Journal* 6 citations 2025
DOI: [10.3847/1538-4357/addabf](https://doi.org/10.3847/1538-4357/addabf)
A. P. Cloonan, G. Khullar, K. A. Napier, M. D. Gladders, H. Dahle, R. Rosener, J. Sullivan, M. B. Bayliss, N. Chicoine, I. Escapa, D. Garza, J. Garza, R. Glusman, K. Gozman, G. Horwath, A. Kisare, B. C. Levine, O. Liang, N. Malagon, M. N. Martinez, A. Masegian, O. S. Matthews Acuña, S. D. Mork, K. Niu, **M. R. Owens**, Y. Pan, J. R. Rigby, K. Sharon, I. Sierra, A. A. Stark, E. Sukay, M. Tagliavia, M. Tamargo-Arizmendi, K. Tavangar, R. Teixeira, K. Tsiane, R. Tu, G. Wagner, E. A. Zaborowski, Y. Zhang, Y. "M." Zhao
- The Sunburst Arc with JWST: II. Observations of an Eta Carinae analog at $z=2.37$,** *Astronomy & Astrophysics* 18 citations 2025
DOI: [10.1051/0004-6361/202450685](https://doi.org/10.1051/0004-6361/202450685)
S. Choe, T. E. Rivera-Thorsen, H. Dahle, K. Sharon, **M. R. Owens**, J. R. Rigby, M. B. Bayliss, M. J. Hayes, T. Hutchison, B. Welch, J. Chisholm, M. D. Gladders, G. Khullar, K. Kim
- The Sunburst Arc with JWST. III. An Abundance of Direct Chemical Abundances,** *The Astrophysical Journal* 30 citations 2025
DOI: [10.3847/1538-4357/ada76c](https://doi.org/10.3847/1538-4357/ada76c)
B. Welch, T. E. Rivera-Thorsen, J. R. Rigby, T. A. Hutchison, G. M. Olivier, D. A. Berg, K. Sharon, H. Dahle, **M. R. Owens**, M. B. Bayliss, G. Khullar, J. Chisholm, M. Hayes, K. J. Kim
- The Sunburst Arc with JWST: I. Detection of Wolf-Rayet stars injecting nitrogen into a low-metallicity, $z=2.37$ proto-globular cluster leaking ionizing photons,** *Astronomy & Astrophysics* 42 citations 2024
DOI: [10.1051/0004-6361/202450359](https://doi.org/10.1051/0004-6361/202450359)
T. E. Rivera-Thorsen, J. Chisholm, B. Welch, J. R. Rigby, T. Hutchison, M. Florian, K. Sharon, S. Choe, H. Dahle, M. B. Bayliss, G. Khullar, M. Gladders, M. Hayes, A. Adamo, **M. R. Owens**, K. Kim
- COOL -LAMPS. VI. Lens Model and New Constraints on the Properties of COOL J1241+2219, a Bright $z=5$ Lyman Break Galaxy and its $z=1$ Cluster Lens,** *The Astrophysical Journal* 4 citations 2024
DOI: [10.3847/1538-4357/ad22de](https://doi.org/10.3847/1538-4357/ad22de)
M. Klein, K. Sharon, K. Napier, M. D. Gladders, G. Khullar, M. Bayliss, H. Dahle, **M. R. Owens**, A. Stark, S. Brownsberger, K. J. Kim, N. Kuchta, G. Mahler, G. Smith, R. Walker, K. Gozman, M. N. Martinez, O. S. Matthews Acuña, K. Merz, J. A. Sanchez, D. J. K. Stein, E. O. Sukay, K. Tavangar
- Resolving Clumpy versus Extended Lya in Strongly Lensed, High-redshift Lya Emitters,** *The Astrophysical Journal* 4 citations 2024
DOI: [10.3847/1538-4357/ad10ad](https://doi.org/10.3847/1538-4357/ad10ad)
A. Navarre, G. Khullar, M. B. Bayliss, H. Dahle, M. Florian, M. Gladders, K. J. Kim, **M. R. Owens**, J. Rigby, J. Roberson, K. Sharon, T. Shibuya, R. Walker
- Small Region, Big Impact: Highly Anisotropic Lyman-continuum Escape from a Compact Starburst Region with Extreme Physical Properties,** *The Astrophysical Journal Letters* 44 citations 2023
DOI: [10.3847/2041-8213/acf0c5](https://doi.org/10.3847/2041-8213/acf0c5)
K. J. Kim, M. B. Bayliss, J. R. Rigby, M. D. Gladders, J. Chisholm, K. Sharon, H. Dahle, T. E. Rivera-Thorsen, M. K. Florian, G. Khullar, G. Mahler, R. Mainali, K. A. Napier, A. Navarre, **M. R. Owens**, J. Roberson

Strongly lensed [O III] emitters at Cosmic Noon with Roman: Characterizing extreme emission line galaxies on star cluster complex scales (100 pc), arXiv 2023

DOI: [10.48550/arXiv.2307.01247](https://doi.org/10.48550/arXiv.2307.01247)

K. J. Kim, M. B. Bayliss, H. Dahle, T. Hutchison, K. Sharon, G. Mahler, **M. R. Owens**, J. E. Rhoads

The Connection between Galactic Outflows and the Escape of Ionizing Photons, The Astrophysical Journal 41 citations 2022

DOI: [10.3847/1538-4357/ac9cd6](https://doi.org/10.3847/1538-4357/ac9cd6)

R. Mainali, J. R. Rigby, J. Chisholm, M. Bayliss, R. Bordoloi, M. D. Gladders, T. E. Rivera-Thorsen, H. Dahle, K. Sharon, M. Florian, D. A. Berg, S. Sharma, **M. R. Owens**, K. Kjellgren, K. J. Kim, J. Wayne

OBSERVATIONS

PRINCIPAL INVESTIGATOR

Resolving the Lyman-alpha kinematics of a lensed galaxy with a rare Lyman-alpha profile, Gemini/GMOS-N 7 hr Gemini 2024B FT

R. Owens, M. Bayliss, E. Solhaug, G. Khullar, B. Welch, K. Kim, M. Gladders, E. Rivera-Thorsen, H. Dahle

Identifying galaxy-lensed Lyman-alpha emitters, 2 hr Gemini 2023A FT Gemini/GMOS-S

R. Owens, K. Kim, M. Bayliss, H. Dahle, J. Burns, K. Sharon, G. Smith, M. Klein, N. Kuchta, R. Walker, E. Rivera-Thorsen, G. Mahler, G. Khullar

COINVESTIGATOR

Spatially Resolving Highly Ionized Channels of Lyman Continuum and Lyman Alpha Escape on 10's of Parsecs Scales In A Strongly Lensed Galaxy, HST/ACS, HST/WFC3 44 orbits HST Cycle 32 GO

M. Bayliss, H. Dahle, M. Florian, M. D. Gladders, T. A. Hutchison, G. Khullar, K. J. Kim, **M. R. Owens**, J. R. Rigby, T. E. Rivera-Thorsen, K. Sharon

Time delay cosmography with strong cluster lenses, HST/ACS, HST/WFC3, JWST/NIRCam 14 orbits, 12 hr HST Cycle 32 GO

H. Dahle, M. Bayliss, A. P. Cloonan, M. Florian, M. D. Gladders, T. A. Hutchison, G. Khullar, K. J. Kim, G. Mahler, K. Napier, **M. R. Owens**, J. R. Rigby, T. E. Rivera-Thorsen, K. Sharon, B. Welch

Spatially Resolved Physical Conditions In a Strongly Lensed X-ray Emitting Dwarf Starburst at Cosmic Noon, Magellan/FIRE 21 hr NOIRLab 2024B

M. Bayliss, J. Roberson, H. Dahle, M. Gladders, G. Khullar, K. Kim, **R. Owens**, K. Sharon

Resolving Multi-Peaked Lyman-alpha In A Strongly Lensed Galaxy at z=4: Revealing A Candidate Highly Magnified LyC Leaking Galaxy, Gemini/GMOS-N 3 hr Gemini 2024A FT

M. Bayliss, A. Navarre, K. Sharon, H. Dahle, **R. Owens**, M. Gladders, K. Kim

A Dynamical Study of an HST+Chandra Sample of Strong Lensing Clusters, Gemini/GMOS-N, Gemini/GMOS-S 28 hr Gemini 2024A

M. Bayliss, R. Gassis, P. Adhikari, M. Gladders, H. Dahle, K. Napier, G. Khullar, G. Mahler, **R. Owens**

The Partially Ionized Medium around HII Regions - the low frequency complement, GBT/PF1 342 MHz 49 hr GBT 23B Regular

P. Salas, K. Emig, **M. Owens**, D. A. Roshi, L. Anderson

Galaxies Under Construction: Resolved Scaling Relations and Stellar Mass Assembly as Revealed by Lensed Star-Forming Clumps at Cosmic Noon, JWST/NIRCam, JWST/NIRSpec 68 hr JWST Cycle 2 GO

M. Florian, R. Bezanson, J. Chisholm, H. Dahle, M. Gladders, T. Hutchison, G. Khullar, K. Kim, G. Mahler, A. Navarre, **R. Owens**, J. Rigby, T. Rivera-Thorsen, J. Roberson, K. Sharon, I. Shivaiei, B. Welch, K. Whitaker

PRESENTATIONS

SCIENCE

The strongly lensed Sunburst Arc galaxy reveals direct Lyman - a escape hundreds of parsecs away from ionizing photon escape, American Astronomical Society meeting #245 (National Harbor, MD, USA) talk 2025

Connecting Lyman - a and ionizing photon escape at cosmic noon with a strongly lensed galaxy arc, American Astronomical Society meeting #243 (New Orleans, LA, USA) talk 2024

Connecting Ly α and LyC escape with a gravitationally lensed, super star cluster at cosmic noon, Escape of Lyman radiation from galactic labyrinths (Kolymbari, Greece) talk 2023

Connecting Ly α and LyC escape with a gravitationally lensed, super star cluster at cosmic noon, Kentucky Area Astronomical Society Meeting 2023 (Lexington, KY, USA) talk 2023

Outflows and Lyman continuum escape in a highly magnified arc, American Astronomical Society meeting #241 (Seattle, WA, USA) poster 2023

Spatially variable Ly α line profiles and environments in a strong LyC leaking galaxy, IAU General Assembly 2022 (Busan, South Korea) poster 2022

Spatially variable Ly α line profiles and environments in a strong LyC leaking galaxy, National Astronomy Meeting 2022 (Coventry, UK) poster 2022

Distribution and dynamics of ionized gas in the Cygnus X massive star-forming complex, American Astronomical Society meeting #240 (Pasadena, CA, USA) poster 2022

Spatially Variable Light and Environments in a Distant Galaxy, University of Cincinnati Undergraduate Scholarly Showcase (Cincinnati, OH, USA) poster 2022

Ultraviolet spectra of the Sunburst Arc, Kentucky Area Astronomical Society Meeting 2022 (Clarksville, TN, USA) talk 2022

Ionized gas in Cygnus X, Exploring Innovation in Appalachia Symposium (online) talk 2022

Ionized gas in Cygnus X, Exploring Innovation in Appalachia Symposium (online) poster 2022

Ionized gas in Cygnus X, 2021 Pulsar Search Collaboratory Capstone Seminar (Morgantown, WV, USA) talk 2021

Ionized gas in Cygnus X, 2021 Pulsar Search Collaboratory Capstone Seminar (Morgantown, WV, USA) poster 2021

Lya emission and gaseous outflows in a $z \sim 2$ lensed galaxy, 2021 Pulsar Search Collaboratory Capstone Seminar (Morgantown, WV, USA) poster 2021

Lya emission and gaseous outflows in a $z \sim 2$ lensed galaxy, University of Cincinnati Undergraduate Scholarly Showcase (Cincinnati, OH, USA) poster 2021

OUTREACH

An Overview of Gravitational Lensing, Astro Night (Berkeley, CA, USA) talk 2025

Our greatest telescopes are billions of light years away, Popping the Science Bubble (Berkeley, CA, USA) talk 2025

TEACHING

Graduate student instructor, Introduction to General Astronomy ASTRON C10 Fall 2025
University of California, Berkeley

Teaching assistant, College Physics II PHYS 2006 Spring 2022
University of Cincinnati

Instructor, Collaborative Learning for PHYS 2006 PHYS 1016 Spring 2022
University of Cincinnati

Teaching assistant, College Physics I PHYS 2005 Fall 2021
University of Cincinnati

Instructor, Collaborative Learning for PHYS 2005 PHYS 1015 Fall 2021
University of Cincinnati

Instructor, Collaborative Learning for PHYS 2006 PHYS 1016 Spring 2021
University of Cincinnati

Teaching assistant, College Physics I PHYS 2005 Fall 2020
University of Cincinnati

Instructor, Collaborative Learning for PHYS 2005 PHYS 1015 Fall 2020
University of Cincinnati

Teaching assistant, College Physics II PHYS 2006 Spring 2020
University of Cincinnati

Instructor, Collaborative Learning for PHYS 2006 PHYS 1016 Spring 2020
University of Cincinnati

Teaching assistant, College Physics I PHYS 2005 Fall 2019
University of Cincinnati

Peer leader, Learning Community MLTI 1011 Fall 2019
University of Cincinnati

SCIENTIFIC WRITING

Is the farthest known star actually a binary system?, Astrobites 2023
R. Owens

How do you find high energy photons when you can't see them?, Astrobites 2023
R. Owens

Are there thousands of Godzillas hiding in plain sight?, Astrobites 2022
R. Owens

REFERENCES

Dan Stark, Professor
University of California, Berkeley

dpstark@berkeley.edu

Loren Anderson, Associate professor
West Virginia University

loren.anderson@mail.wvu.edu

Matthew Bayliss, Associate professor
University of Cincinnati

baylismb@ucmail.uc.edu

Alexandre Sousa, Professor
University of Cincinnati

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